ChestAI RADIOLOGY PLATFORM

Deep Learning Diagnostic & Explainability Consultation Report

REPORT ID: CAI-2610042355
DATE: 2026-10-04 23:55
STATUS: PRE-READ COMPLETE

Patient MRN: CLI-PT-001	**Age / Sex:** 65 y/o / Unspecified	**Projection:** PA (Posteroanterior)
Indication: CLI automated headless diagnostic evaluation.	**Sensitivity Threshold:** 50%	**Triage:** CRITICAL ALERT · IMMEDIATE REVIEW

QUANTITATIVE PATHOLOGY ASSESSMENT

Target Pathology	Predicted Probability	Cutoff	Classification Result	Calibration
Cardiomegaly	**100.0%**	50%	**POSITIVE (Detected)**	Temperature Scaled
Pleural Effusion	**0.0%**	50%	**NEGATIVE (Clear)**	Temperature Scaled
Atelectasis	**0.0%**	50%	**NEGATIVE (Clear)**	Temperature Scaled
Edema	**100.0%**	50%	**POSITIVE (Detected)**	Temperature Scaled

IMAGING & GRAD-CAM EXPLAINABILITY ATTENTION

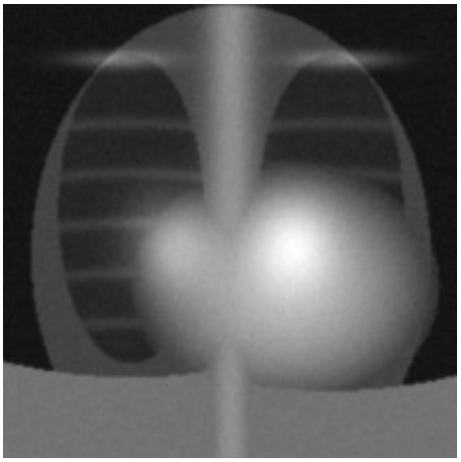

Figure 1: *Original Chest Radiograph*

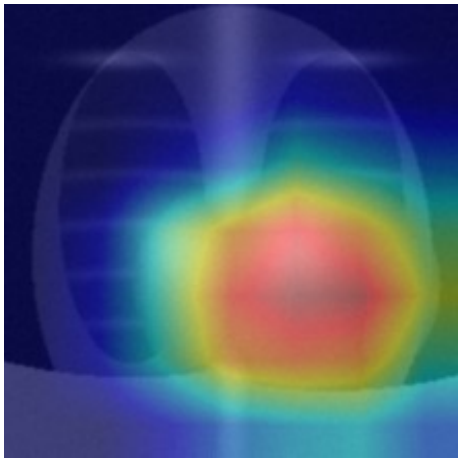

Figure 2: *Grad-CAM Heatmap Attention Overlay*

STRUCTURED IMPRESSION & CLINICAL ACTION PLAN

Impression:	Radiograph exhibits salient features consistent with **Cardiomegaly, Edema** above the specified operating threshold (50%). Attention heatmaps demonstrate corresponding anatomical activation.
Action Plan:	• • Recommend immediate clinical correlation with oxygenation status and consideration of urgent diuretic therapy or thoracic ultrasound. • Echocardiographic assessment recommended to evaluate left ventricular ejection fraction and valvular anatomy.

CLINICAL DISCLAIMER & RESEARCH NOTICE: *ChestAI is an investigational deep learning research platform. Outputs are probabilistic assistive recommendations and do not constitute an official medical diagnosis. All interpretations must be validated by a credentialed radiologist before altering patient management.*